MODUL UNIVERSITY VIENNA

SEID 2364: Societal and Ethical Impact of Data Science

SESSION 3

Harm

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We are going to take the education adventure together. We are going to marco, and go micro, and we are going to see why MU is the right place for you. Lets begin.

Opening Provocations

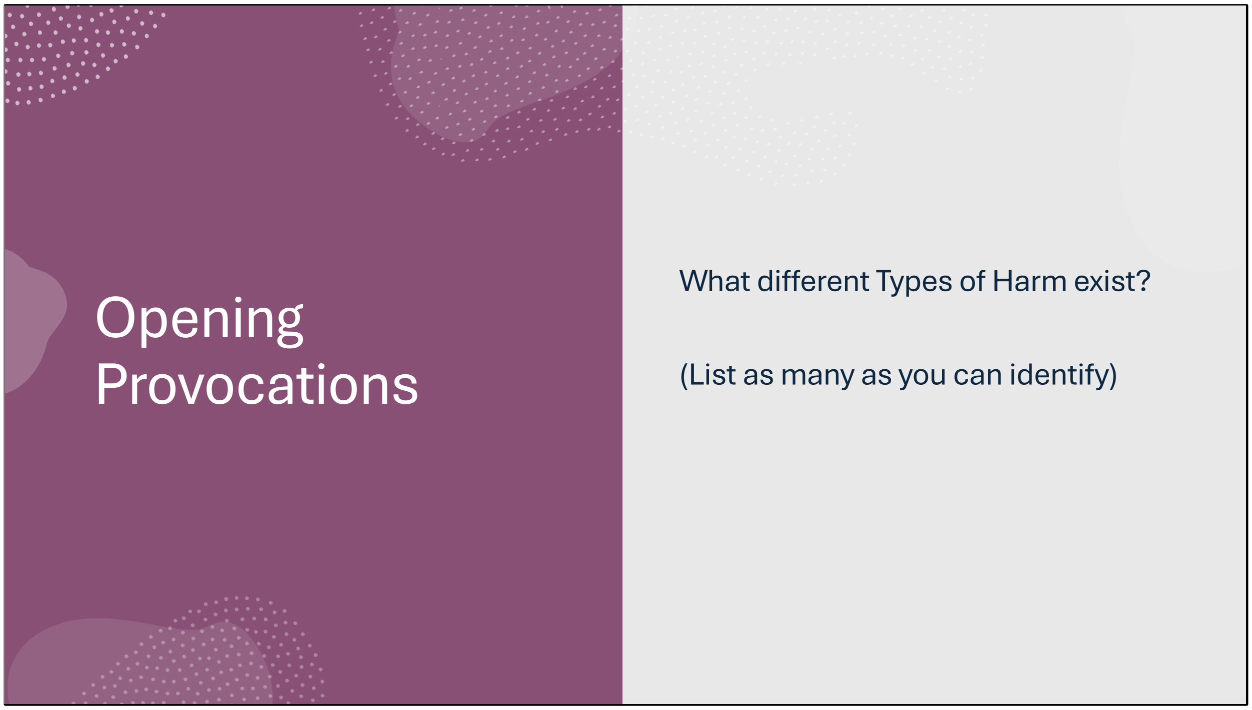
- Write in your private journal
- Stream of consciousness.
- Capture your **immediate intuition**, not a polished or argued answer.
- Do not analyze or justify.
- Words or short phrases are enough.

(Write. Do not discuss yet.)

Opening Provocations

What is “Harm”?

(Write. Do not discuss yet.)



(Write. Do not discuss yet.)

Opening Provocations

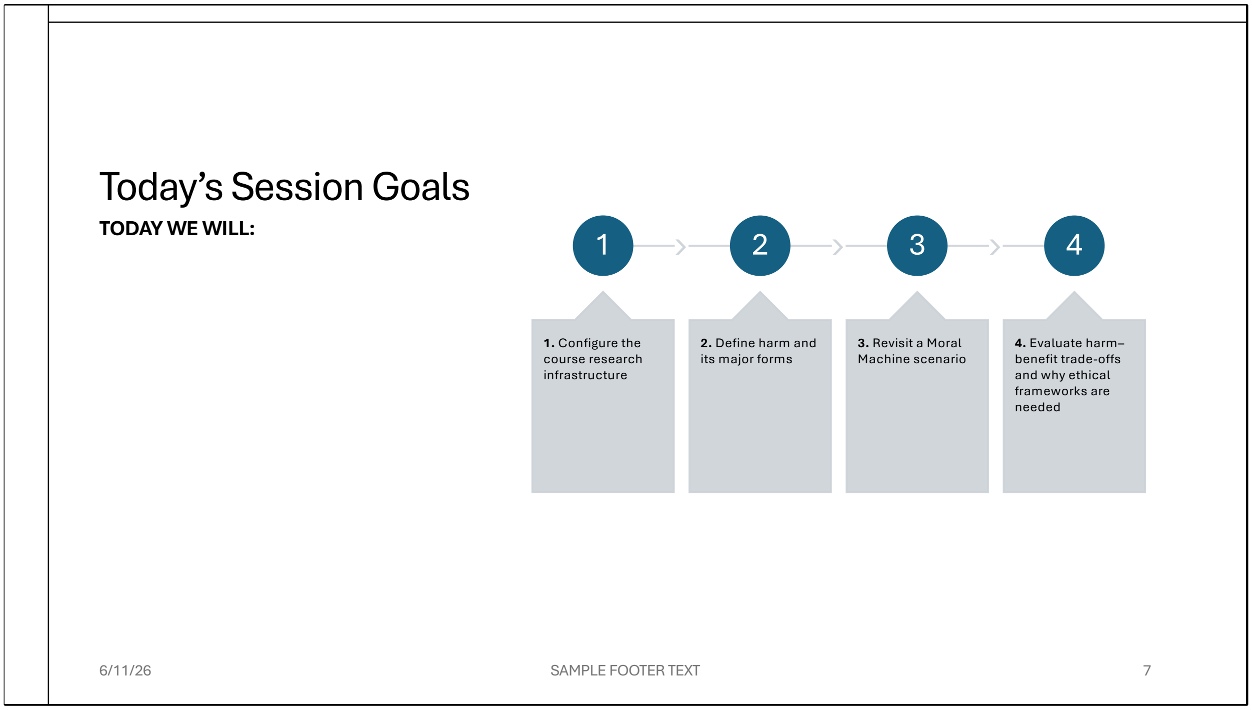
Can Harm be measured?

Explain your reasoning.

(Write. Do not discuss yet.)

SESSION OVERVIEW & SETUP

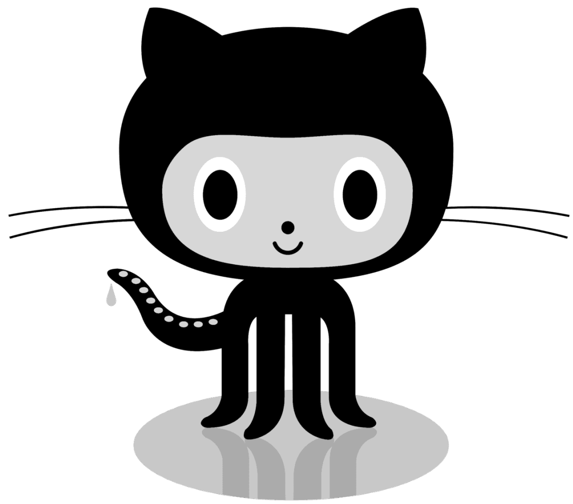
Today's Session Goals, GitHub Setup Instructions, Student Repository Structure, Zotero Setup and Verification



Today we will:

- Configure the course research infrastructure
- Discuss harm and different forms of harm
- Revisit a Moral Machine scenario
- Examine harm–benefit trade-offs
- Introduce the need for ethical frameworks

GITHUB Setup



COURSE REPO: CHI-CITYTECH/SEID-2364

https://github.com/CHI-CityTech/SEID-2364

1. Open GitHub Desktop
2. Clone the repository
3. Create a branch: lastname-firstname
4. Create your folder in /students

Course Repository
https://github.com/CHI-CityTech/SEID-2364
Steps:
Open GitHub Desktop
Clone the repository
Create a personal branch
Create your folder inside the students directory
Branch name format:
lastname-firstname

Image source: Microsoft 365 content library

REPO STRUCTURE

Setup for next assignment

1) Create **/students/lastname-firstname** in the repository.
2) Add file **harm-analysis.md**.
3) Commit and push to your branch (used next).

Inside the repository create:
/students/lastname-firstname
Create file:
harm-analysis.md
Commit and push to your branch.
This will be used for your next assignment.

ZOTERO SETUP CHECK

INSTALL

• Zotero Desktop
• Zotero Browser Connector

VERIFY

• Capture a source from your browser
• Confirm it appears in your Zotero library

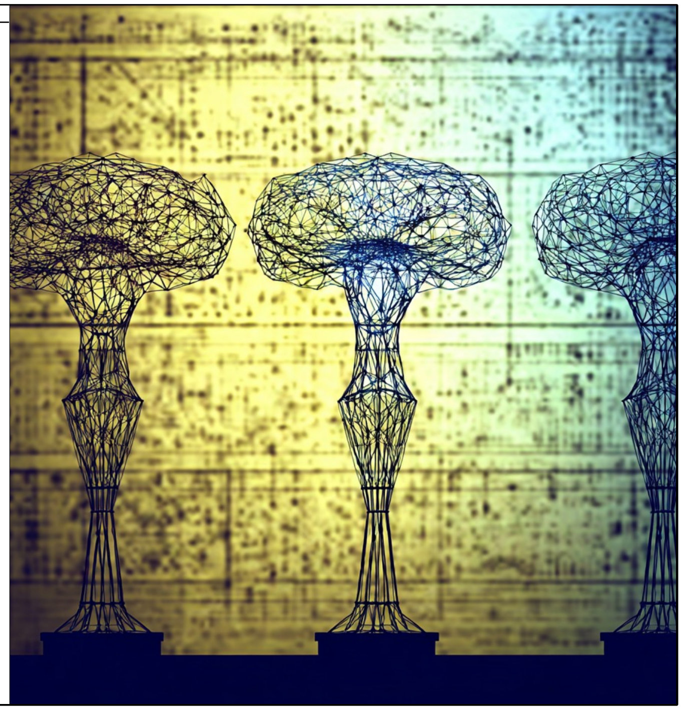
Zotero is the shared research library for this course.

Install:
• Zotero Desktop
• Zotero Browser Connector
Verify:
• You can capture a source from your browser
• The reference appears in your Zotero library
Zotero will serve as the shared research library for this course.

Image source: Microsoft 365 content library

Class Activity – Intelligence Models

- Open the **Google Doc** (link in chat)
- Duplicate the **Template tab**
- Rename the tab with **your name**
- Paste:
- your definition of intelligence
- your rule set
- We will compare models across the class.
- Exercise 1: Mapping Intelligence



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Discussion – Comparing Intelligence Models

What **rules or conditions** define intelligence?

Do we **agree or disagree** with these rules?

Do the rules work for **both human and machine intelligence**?

Where is the **boundary between non-intelligent and minimally intelligent systems**?

Do the rules produce results that **conflict with our intuition**?

Are there **edge cases where the rules break down**?

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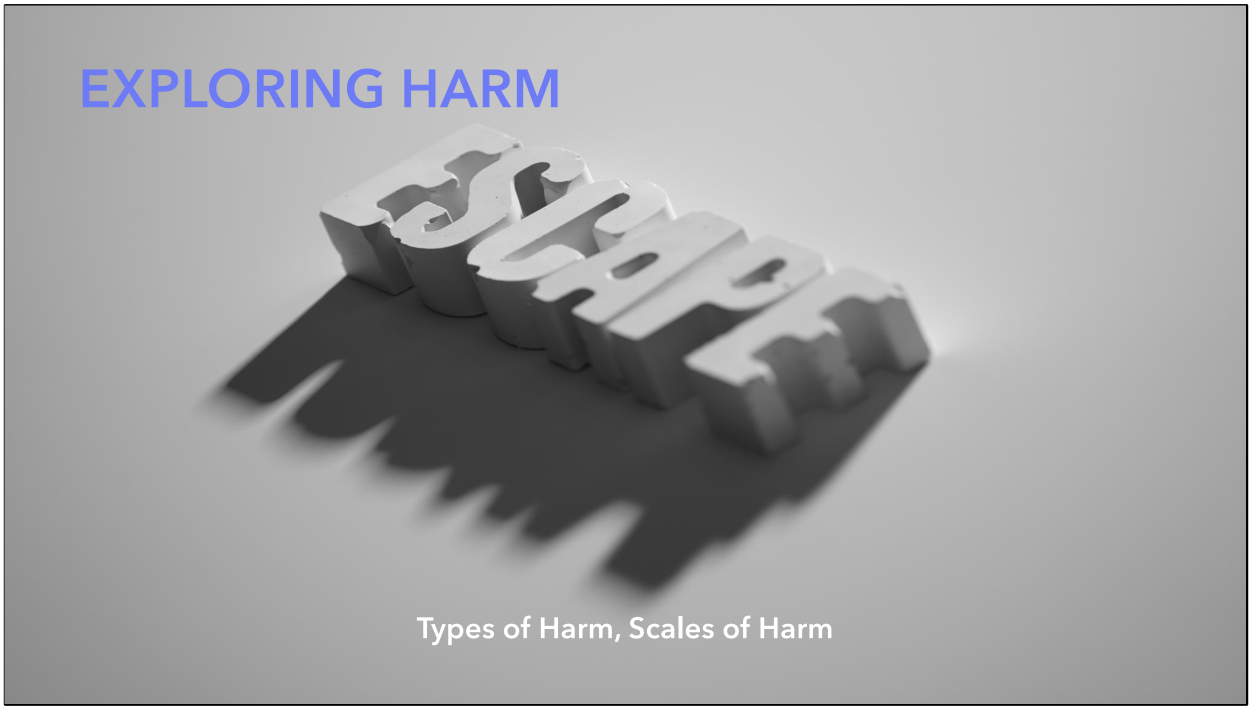
Discussion – Comparing Intelligence Models
After reviewing several student models, consider the following questions:
Which **rules or conditions** appear across multiple definitions?
Which models require **learning**, **adaptation**, or **goal-directed behavior**?
Could the **same system** be classified differently under different rule sets?
What happens if a system that satisfies these rules **acts in the world**?
Key idea
Definitions of intelligence function as **analytical frameworks** that classify systems.
If intelligence leads to **action**, actions produce **consequences**.
Those consequences may involve **harm or benefit**, which raises the question:
How should societies evaluate those outcomes?



Types of Harm Identified, Scales of Harm

Image source: Microsoft 365 content library

What types of harm did you identify?
Examples:
- Physical harm
- Psychological harm
- Economic harm
- Reputational harm
- Social harm
- Institutional harm

Image source: Microsoft 365 content library

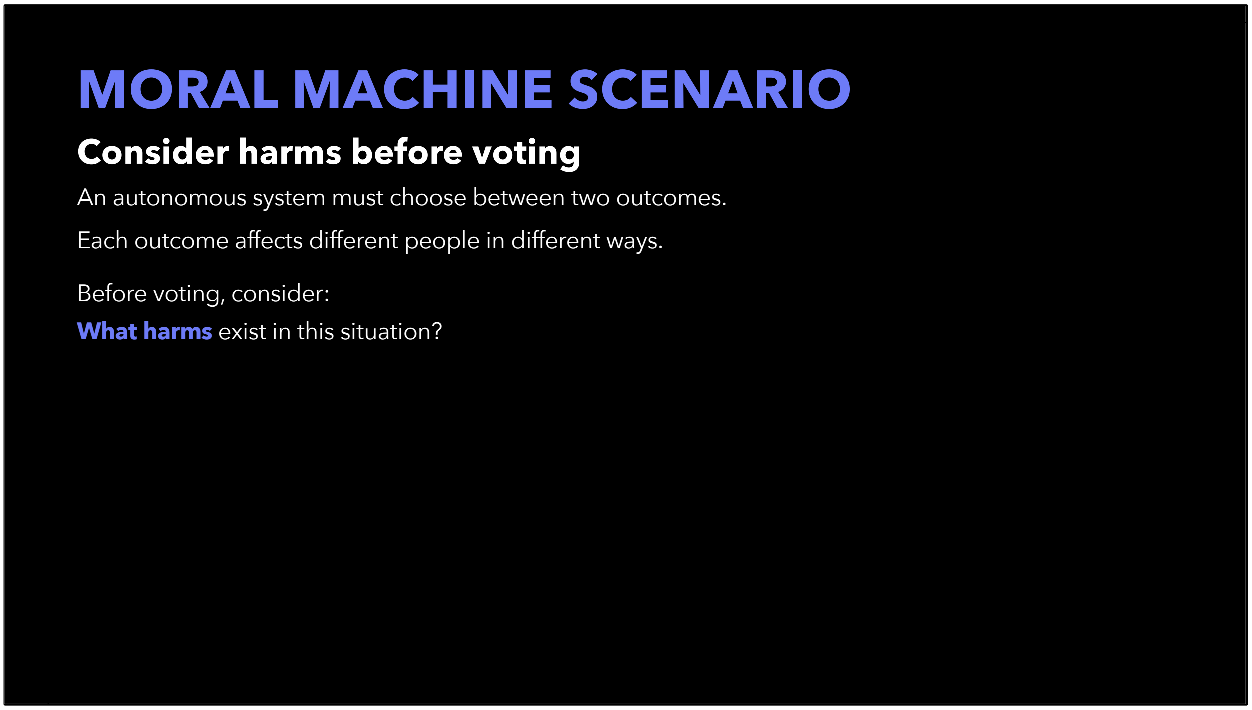
Harm can occur at multiple levels:
- Individual
- Group
- Organization
- Society

A single action may produce harms across multiple levels.

Image source: Microsoft 365 content library

Moral Machine Scenario, Are Any Choices Harm-Free?

Image source: Microsoft 365 content library



Consider the following decision scenario.
An autonomous system must choose between two outcomes.
Each outcome affects different people in different ways.
Before voting, consider:
What harms exist in this situation?

HARM-FREE CHOICES?

In many real decisions, no option completely eliminates harm.
Instead, choices redistribute harm across possible outcomes.

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No option completely eliminates harm.
Instead, decisions redistribute harm across possible outcomes.

Image source: Microsoft 365 content library

HARM–BENEFIT TRADE-OFFS

Harm and benefit examples: the ethical problem

Harm and Benefit Examples, The Ethical Problem

Many actions produce both harm and benefit.
Example:
Action A
• Benefits group X
• Harms group Y
Action B
• Benefits group Y
• Harms group X

Image source: Microsoft 365 content library

THE ETHICAL PROBLEM

Choosing among harms and benefits

If multiple actions produce different combinations of harm and benefit:

How do we decide?

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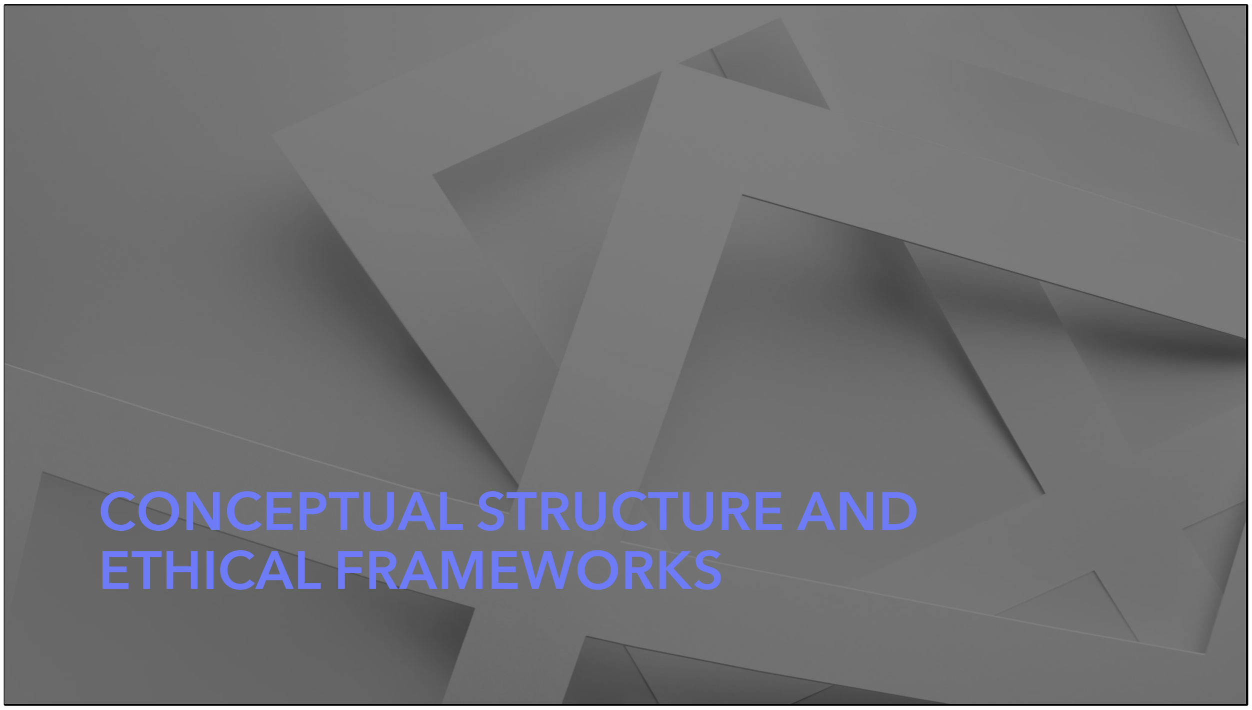

Conceptual Structure of Harm and Ethics, Why Ethical Frameworks Exist

Image source: Microsoft 365 content library

Intelligence → produces actions
Actions → create consequences
Consequences → produce harm or benefit
Ethics → evaluates those outcomes

Image source: Microsoft 365 content library

WHY FRAMEWORKS EXIST

Ethical frameworks provide methods for evaluating:

• competing harms • competing benefits • trade-offs between outcomes

We will begin studying these frameworks next.

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Assignment Overview, Closing Question, Next Class Preview

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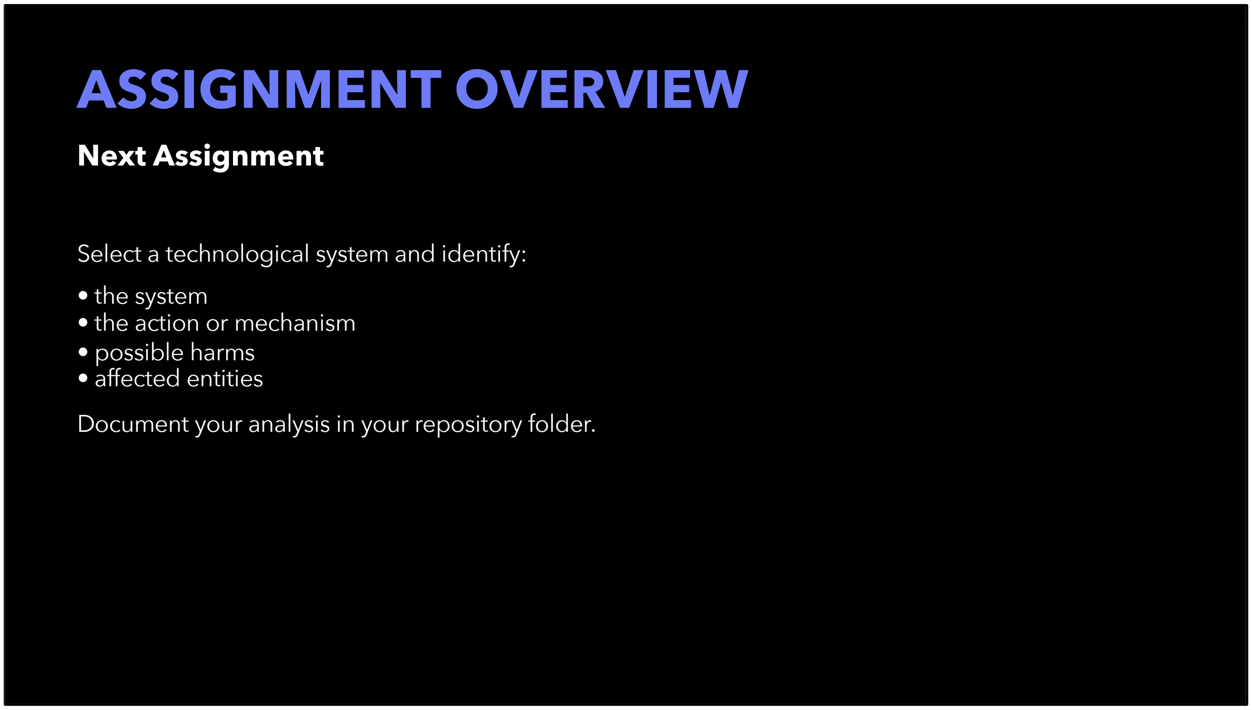

Next Assignment
Select a technological system and identify:
- the system
- the action or mechanism
- possible harms
- affected entities

Document your analysis in your repository folder.

CLOSING QUESTION

IF INTELLIGENT SYSTEMS CAN PRODUCE ACTIONS THAT CAUSE HAR
WHO SHOULD DECIDE HOW THOSE HARMS ARE EVALUATED?

If intelligent systems can produce actions that cause harm,
who should decide how those harms are evaluated?

NEXT CLASS PREVIEW

We’ll begin examining ethical frameworks and how they’re used to evaluate decisions in complex systems.

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